Numerical Analysis of the Faddeeva Function $w(z)$: Accuracy Mapping and High-Precision Computation of Zeros

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This report evaluates the numerical reliability of the Faddeeva function w(z) within the `libcerf` library. We pinpoint significant precision losses in the lower half-plane caused by catastrophic cancellation near the function's roots. Using `python-flint` as a high-precision benchmark, we refine these zeros to 24-digit accuracy through an asymptotic-Newton scheme to analyze and mitigate these instabilities.

1 Methodology for Accuracy Validation

To validate the accuracy of `libcerf`, we compare its double-precision output with a high-precision reference computed using the Arb library, accessed through `python-flint`. For a given complex argument z, let $w_{\text{libcerf}}(z)$ denote the value returned by `libcerf` and $w_{\text{ref}}(z)$ the high-precision reference. We define the total relative error (TRE) as

$$\text{TRE}(z) = \frac{\left|w_{\text{libcerf}}(z) - w_{\text{ref}}(z)\right|}{\left|w_{\text{ref}}(z)\right|}. \tag{1}$$

The reference values are evaluated with at least 96 bits of precision so that the numerical uncertainty of $w_{\text{ref}}(z)$ remains well below double-precision machine epsilon ($\epsilon \approx 2.22 \times 10^{-16}$).

2 Identification of Numerical Inaccuracies

In agreement with [2], the implementation in `libcerf` exhibits near-optimal accuracy in the upper half-plane ($\operatorname{Im} z \geq 0$), while the lower half-plane ($\operatorname{Im} z < 0$) is more delicate because values there are obtained via a reflection formula involving $w(\bar{z})$ and an exponential term.

2.1 Detection of Catastrophic Cancellation

Using a scan script, we evaluate $w_{\text{libcerf}}(z)$ and $w_{\text{ref}}(z)$ in the vicinity of theoretically predicted zeros. The results, normalized by the machine epsilon (ϵ), are summarized in Table 1.

Table 1: Numerical scan results for $w(z)$ accuracy.

Test Region	Coordinate $z = x + iy$	Relative Error (TRE)
Upper Half-Plane ($y > 0$)	$9.01 + 0.12i$	$\approx 0.01\epsilon$
Near First Zero ($y < 0$)	$1.99 - 1.35i$	1.18×10^{-8}
Lower Half-Plane ($y < 0$)	$4.83 - 1.60i$	≈ 0.00

This behavior is consistent with catastrophic cancellation inside the reflection formula, where an algebraic tail and an exponentially small term cancel almost completely near a root of $w(z)$.

3 High-Precision Determination of Zeros

To obtain reliable reference values for the zeros of $w(z)$, we developed a tool `compute_zeros.py` that computes complex zeros z_n to a relative tolerance of 10^{-24} using Arb.

3.1 Asymptotic Initial Approximation

The zeros of $w(z)$ are directly related to the zeros of the complementary error function $\operatorname{erfc}(z)$ via $w(z) = \mathrm{e}^{-z^2} \operatorname{erfc}(-iz)$. Asymptotic expansions for the zeros of $\operatorname{erfc}(z)$ are given in DLMF §7.13(ii). For the nth zero, we introduce

$$\lambda = \sqrt{\left(n - \tfrac{1}{8}\right)\pi}, \qquad \mu = \ln\left(2\lambda\sqrt{2\pi}\right), \tag{2}$$

and approximate the coordinates (x_n, y_n) in the erfc-plane by [1]

$$x_n \sim -\lambda + \frac{\mu}{4}\lambda^{-1} - \frac{1}{16}\left(1 - \mu + \tfrac{1}{2}\mu^2\right)\lambda^{-3}, \tag{3}$$

$$y_n \sim \lambda + \frac{\mu}{4}\lambda^{-1} + \frac{1}{16}\left(1 - \mu + \tfrac{1}{2}\mu^2\right)\lambda^{-3}. \tag{4}$$

The corresponding zero of the Faddeeva function is then obtained by the rotation

$$z_{w,n}^{(0)} = y_n + i x_n, \tag{5}$$

which serves as an initial guess for the refinement of the nth zero of $w(z)$.

3.2 Newton–Raphson Refinement

Starting from the asymptotic estimate $z_{w,n}^{(0)}$, we apply the Newton–Raphson method to the Faddeeva function:

$$z_{k+1} = z_k - \frac{w(z_k)}{w'(z_k)}. \tag{6}$$

The derivative of $w(z)$ can be expressed in closed form as

$$w'(z) = -2z\, w(z) + \frac{2i}{\sqrt{\pi}}, \tag{7}$$

which allows an efficient high-precision evaluation within Arb.

3.3 Convergence and Accuracy Criterion

The iteration is stopped once the step size satisfies the relative tolerance

$$|\Delta z| < |z| \cdot 10^{-24}. \tag{8}$$

In the Arb implementation, this condition is enforced using interval bounds: the step size is overestimated via `abs_upper()` and the magnitude $|z|$ is underestimated via `abs_lower()`, yielding a conservative guarantee for the final error.

4 Asymptotic Expansion and Reflection Principle

In order to assess the numerical stability of $w(z)$ in the lower half-plane, we rely on the asymptotic expansion of the Faddeeva function. For $|z|$ sufficiently large, an asymptotic representation of the form

$$w(z) \sim \frac{i}{\sqrt{\pi}\, z}\left(1 + \sum_{m=1}^{N} \frac{\left(\frac{1}{2}\right)^{\overline{m}}}{z^{2m}}\right) + \delta\, 2\mathrm{e}^{-z^2} \tag{9}$$

where $\left(\frac{1}{2}\right)^{\overline{m}}$ denotes Knuth's raising power notation.

Near the zeros z_n, the algebraic tail and the exponential reflection term can be of comparable magnitude but opposite sign, so that small perturbations in either contribution may lead to a large relative error in their sum. This mechanism explains the sensitivity of the reflection formula in the vicinity of roots and motivates a more careful treatment in these regions.

5 High-Precision Zeros as Reference Data

The columns in Table 2 provide a detailed breakdown of the numerical behavior of the Faddeeva function near its roots:

- n **and Coordinates**: The first three columns identify the zero index and its high-precision coordinates ($z_n = x_n + iy_n$), refined via the Newton–Raphson scheme to a target relative accuracy of 10^{-24}.
- **Agreement Digits ($N = 1, 2, 3$)**: These columns quantify the accuracy of the asymptotic expansion for different truncation orders. Since the theoretical value at these points is $w(z_n) = 0$, the agreement digits are defined as:

$$\text{Digits} = -\log_{10} |w_{\text{asy}}(z_n)| \tag{10}$$

  This metric represents the number of leading zeros in the residual. For instance, an agreement of 2.28 for z_1 at $N = 1$ implies a residual magnitude of approximately $10^{-2.28} \approx 0.005$.

The data reveals that although the accuracy improves as N increases and as the zeros move further into the far-field (higher $|z|$), the agreement remains significantly below the 16-digit threshold required for double-precision stability. This confirms that the reflection formula, while mathematically correct, is numerically insufficient near the roots of the function.

These values provide high-quality reference points for testing both the reflection formula and any local expansions designed to stabilize the evaluation of $w(z)$ near its zeros.

Table 2: Refined zeros z_n with 24-digit accuracy and corresponding agreement digits $(-\log_{10}|w_{\text{asy}}|)$ for orders $N = 1, 2, 3$.

n	$\text{Re}(z_n)$	$\text{Im}(z_n)$	$N = 1$	$N = 2$	$N = 3$
1	1.99146684283387957728216	-1.35481012811200624889985	2.28	2.66	2.90
2	2.69114902425143882885089	-2.17704490608961591153926	3.07	3.75	4.29
3	3.23533086835281645004558	-2.78438761323042816881586	3.52	4.39	5.11
4	3.69730970246846839780755	-3.28741078938984856745678	3.84	4.84	5.68
5	4.10610728468263205492209	-3.72594871944579042386240	4.09	5.18	6.13
6	4.47681569296754570200572	-4.11963522761173052587081	4.29	5.46	6.49
7	4.81848829188331916543408	-4.47983279773120232166478	4.46	5.70	6.80
8	5.13706727126634745718011	-4.81380668204443425449946	4.61	5.91	7.06
9	5.43670391073399739807528	-5.12653154549691946318386	4.74	6.09	7.29
10	5.72043485101455238606114	-5.42158857692298128694983	4.86	6.25	7.50

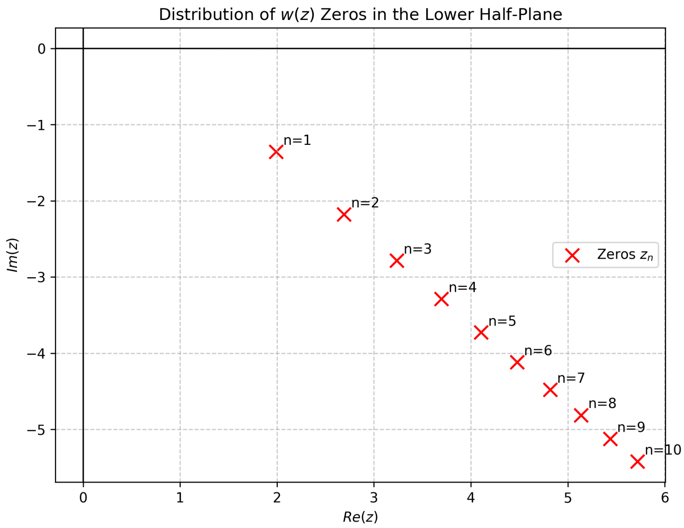

Figure 1: Location of the first ten zeros of $w(z)$ in the complex plane.

6 Perspective: Local Taylor Expansions Near Zeros

A possible strategy to avoid catastrophic cancellation in a production library is to use local Taylor expansions around precomputed zeros in the problematic region $\text{Im}\, z < 0$, similar to the Taylor-based approach used in the first quadrant in [2]. Near a zero z_n, one can write

$$w(z) \approx \frac{2i}{\sqrt{\pi}}(z - z_n) - \frac{2iz_n}{\sqrt{\pi}}(z - z_n)^2 + \mathcal{O}\big((z - z_n)^3\big), \tag{11}$$

which avoids the explicit subtraction of large nearly equal terms that appears in the reflection formula. Combined with tabulated high-precision zeros such as those in Table **??**, this approach offers a principled way to "bypass" the most sensitive regions in the lower half-plane if such a modification is integrated into `libcerf`.

References

[1] J. Wuttke. The Faddeeva function $w(z)$. Manuscript rff0, Forschungszentrum Jülich, 2025.

[2] J. Wuttke. Code generation for computing an analytical function with near machine precision on square tiles, with application to the Faddeeva function. Manuscript cgt0, Forschungszentrum Jülich, 2025.